

The Method of Exhaustion

How Ancient Greek Geometry Bounded the Infinite

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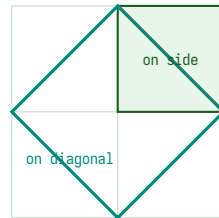
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A Number That Was Not a Number

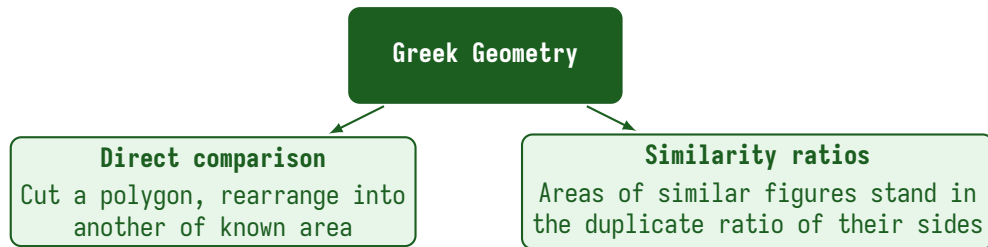
A Pythagorean drew the diagonal of a square and upon it raised a second square which covered twice the first.

Does a single measure then divide both side and diagonal, each a whole number of times? Any such measure forces a smaller, and smaller still, without end.



twice the square on the side

And this was the *easy* case. The circle posed a deeper problem. Its boundary is curved, its area is not polygonal, and its ratio of circumference to diameter would later prove to be no ratio of integers at all. How, then, could Greek geometry prove exact statements about circular area without treating infinity as an object?



Both are finite. Neither works for a circle or a sphere.

Antiphon the Sophist (480–411 BCE), reported method

Antiphon inscribed regular polygons in a circle, repeatedly doubling the number of sides, arguing that the remaining gap would eventually be exhausted

So "eventually equals" is an assertion about infinity and infinity is not an object of proof. **The real problem was proof-theoretic**, not numerical.

Eudoxus of Cnidus (390–337 BCE) answered with a new definition of equality of ratios of magnitudes

Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρὸς δεύτερον καὶ τρίτον πρὸς τέταρτον, ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκεις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκεις πολλαπλασίων καθ' ὅποιονοῦν πολλαπλασιασμὸν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέχη ἢ ἅμα ἴσα ἢ ἢ ἅμα ἐλλείπη ληφθέντα κατάλληλα.

Or in modern notation: $a : b = c : d$ iff for all positive integers m, n

$$ma > nb \iff mc > nd, \quad ma = nb \iff mc = nd$$

Two ratios are equal iff they agree on *every* rational test. The ratio is defined by its behaviour against all rationals no numerical value is ever produced.

Modern echo. This is Dedekind's construction of the real numbers (1872 CE). Eudoxus anticipated it by 2200 years.

Controlling the Infinite With the Finite

One technical lemma does the heavy lifting. The *Archimedean halving property*, due to Eudoxus:

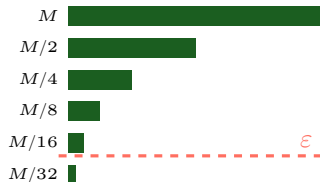
Elements, Book X, Proposition 1

If from any magnitude there be subtracted a part not less than its half, and from the remainder a part not less than its half, and so on continually, there will at length remain a magnitude less than any assigned magnitude of the same kind

Modern restatement: for every $\varepsilon > 0$ and every magnitude M , there exists n with $(\frac{1}{2})^n M < \varepsilon$. Finite. No “as $n \rightarrow \infty$.”

Rather: **for every tolerance, eventually.**

Modern echo. The ε - δ discipline of Cauchy (1821 CE) and Weierstrass, 2300 years early.



The Exhaustion Schema

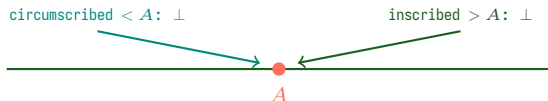
To prove the area of a curved figure equals A rule out both strict inequalities.

If area $> A$

Inscribe polygons, halving the uncovered region each step, until the remainder falls below A . The inscribed polygon then exceeds A which is impossible

If area $< A$

Circumscribe polygons, their excess shrinks below A . The circumscribed polygon then falls below A . Same contradiction, reversed



Therefore the area equals A . **No limit is ever taken.**

Modern echo. Riemann integration: $\underline{S}_n \leq \int f \leq \overline{S}_n$, and the gap shrinks.

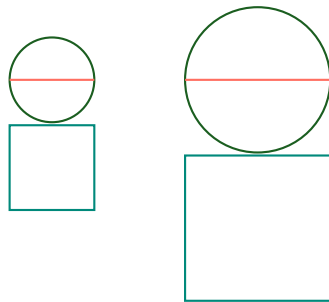
Elements, Book XII, Proposition 2

Circles are to one another as the squares on their diameters

Notice what this result *is* and *is not*

- **Not** a value of π . No number for circular area ever appears.
- **A structural law:** circles behave *relative to each other*, scaling as squares on any linear dimension.

The constant that relates a circle to a square is a separate, harder question left to Archimedes.



circles :: squares on diameters

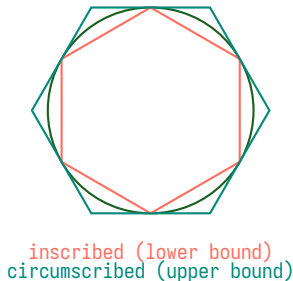
Measurement of a Circle, Proposition 3 (250 BCE)

The perimeter of every circle exceeds three times the diameter by less than one-seventh, and by more than ten seventy-oneths, of the diameter

Archimedes applied exhaustion to a 96-gon (six doublings from a hexagon) bounding the ratio above and below.

Modern restatement: $3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}$

Four-digit accuracy with a *proven enclosure* it is not even a measurement, a theorem about the ratio location.



What Archimedes Did Not Publish

Archimedes sent Eratosthenes a private letter, *The Method of Mechanical Theorems*, recovered in 1906 by J.L. Heiberg from a Byzantine palimpsest overwritten with a prayer book.

Procedure. Slice a figure into parallel sections, treat each as an *indivisible* of definite weight, hang them on a lever opposite sections of known magnitude. Arms balance, and fulcrum distances deliver the unknown. So the sphere is two-thirds its circumscribing cylinder, the parabolic segment four-thirds its inscribed triangle.

“It is easier to find the proof once one has acquired some knowledge of the questions by the method.” Archimedes to Eratosthenes

Neither result was published in this form. Each was re-derived by exhaustion first. Indivisibles for discovery, double-contradiction for proof.

Quadrature of the Parabola

The area of a parabolic segment equals $\frac{4}{3}$ of its inscribed triangle T

Modern view

$$T\left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right) = \frac{4}{3}T.$$

Archimedes' view

for every n , he filled the segment with triangles of total area

$$T\left(1 + \frac{1}{4} + \dots + \frac{1}{4^n}\right) \quad \text{with remainder at most} \quad \frac{1}{3} \cdot \frac{1}{4^n} T.$$

Archimedes never summed an infinite series. He wrote a *finite identity* together with a *remainder that can be driven below any magnitude*. The two together pin the answer, and infinity never enters the proof.

Modern echo. A partial sum with a shrinking remainder bound is still how we prove convergence today.

Three Principles Behind the Method

Three principles of Greek mathematics made the whole machinery work.

- 1 **Magnitude over number.** The primary objects were ratios of geometric magnitudes, not decimal values. An irrational was not a defect to be removed but a kind of thing to be handled.
- 2 **Proof over computation.** An approximation without a certified error bound counted for nothing. A bound with no computed value counted as a theorem.
- 3 **Finite control over infinite process.** A shrinking sequence was handled through the law it obeys, never through the limit point it approaches.

The method of exhaustion is those three principles turned procedural. It is inseparable from the philosophy of proof that produced it.

The schema sits at the operational core of modern mathematics.

- **Dedekind real numbers.** A direct formalisation of Eudoxus' ratios.
- **ε - δ analysis** (Cauchy, Weierstrass). The Archimedean lemma turned into the definition of a limit.
- **Riemann integration.** Lower and upper sums bracket the curved region, and the gap is driven to nothing.
- **Interval arithmetic and verified numerics.** Archimedes' two-sided enclosure, now standard in computer-assisted proofs.
- **A posteriori error estimates in FEM.** A computed solution ships with a rigorous bound on its own error, and the mesh is refined wherever that bound is largest.

In every case the ancient pattern survives. Approximate by simpler objects, then *prove the error shrinks*.

The method of exhaustion is often called “ancient integral calculus.” That label is too loose and too modest.

The Greek breakthrough was not a calculus algorithm. It was a **proof discipline for infinite processes that never admits infinity as an object.**

Approximate, bound, contradict. That triple still runs modern numerical analysis, and its echo is heard throughout real analysis. Greek mathematics did not produce *numbers* for curves. They produced *ratios* and *proofs*. The numbers arrived later. The reasoning is still ours.

Thanks for your attention